

CS-334: Machine Learning

Lecture 09: Model Assessment

09/24/2026

Prof. Shengpu Tang



Announcements

HW2 - due Sunday

Topics covered

- Decision Boundaries (Lec #3)
- Perceptron (Lec #4)
- GD, SGD (Lec #5)
- Linear Regression (Lec #6)
- Regularization (Lec #7)

Solutions to be released Wed

Grades: Thursday or next week

HW3 – will be released tomorrow

Topics covered

- Feature Engineering (Lec #2, #11)
- Logistic Regression (Lec #8)
- Model Assessment (Lec #9)
- Model Selection (Lec #10)
- Feature Selection (Lec #11)

Due in two weeks Sun 11:59pm

Extra Credit due Wed 10/14 11:59pm

- The ROC curve (Lec #9)

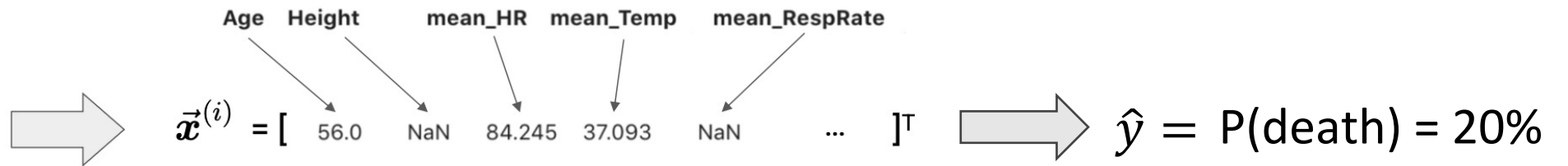
Homework 3

“Predicting Survival of ICU Patients”



Goal: based on data collected during the first 48 hours of the ICU admission, predict chance of survival

Time	Variable	Value
00:00	Age	56.00
00:00	Gender	0.00
00:00	Height	-1.00
00:00	ICUType	4.00
00:00	Weight	48.60
	⋮	
47:16	Urine	180.0
47:31	DiasABP	89.0
47:31	HR	121.0
47:31	MAP	114.0
47:31	SysABP	158.0



HW3 - grading

[80 pt] Guided process of building a full ML pipeline

[18 pt] autograded code implementation

[62 pt] written answers

[20 pt] Open-ended challenge.

There is a set of **unseen** patients (you only have features) on which that your model will be evaluated.

[10 pt] written answers. Describe what you did

[10 pt] model performance. **Graded on prediction quality!**

Problem Setup

- Classification
- Regression

minimize regularized empirical risk

loss function

- ▶ zero-one loss
- ▶ perceptron loss
- ▶ logistic loss
- ▶ hinge loss
- ▶ squared loss

regularization

- ▶ L2
- ▶ L1
- ▶ ElasticNet

Trade-off: bias vs. variance

Supervised Learning: $X \rightarrow Y$

Goal: generalization to unseen data

Optimization tools

- stochastic gradient descent
- Newton's method*
- closed-form solution
- Lagrangian dual*

Linear Models

Perceptron

Logistic regression

SVM*

Ridge regression

LASSO

Supervised Learning : $X \rightarrow Y$

Goal : generalization to unseen data

Practical Aspects

- model assessment
- model selection
- feature selection
- interpretability

Non-linear Models

- $\phi(\vec{x})$ explicit feature mapping
 - $K(\vec{x}^{(i)}, \vec{x}^{(j)})$ kernels
 - decision tree & random forest (bagging)
 - boosting
↳ Adaboost
 - neural networks
 - feed-forward
 - convolutional
 - recurrent
 - transformers
- ensemble methods
- nonlinear activation
 - regularization techniques

Unsupervised Learning : X

Goal : representation characterization

Clustering

- partitional: k-means
- hierarchical*
- spectral*

Recommender Systems

- Collaborative filtering
- nearest neighbors
 - matrix factorization

Autoencoders

Reinforcement Learning

agent wants to achieve some goal

Dilemma:
exploration
vs
exploitation

Multi-Armed Bandit

ϵ -greedy --- ϵ
optimistic initialization --- Q_0
upper confidence bound --- C

Linear Regression Model Implementation

<code>sklearn.linear_model.LinearRegression</code>	wrapper of <code>scipy.linalg.lstsq</code> , uses SVD
<code>sklearn.linear_model.Ridge</code>	can select solver (SGD variants, SVD, etc)
<code>sklearn.linear_model.Lasso</code>	} Lasso is just ElasticNet with L2-penalty “turned off” Implemented as coordinate descent
<code>sklearn.linear_model.ElasticNet</code>	
<code>sklearn.linear_model.SGDRegressor</code>	Generic SGD routine, mix and match different loss and regularizer

Linear Classification Model Implementation

<code>sklearn.linear_model.LogisticRegression</code>	L1/L2/elasticnet regularized logistic regression, $C = 1/\lambda$ is inverse of regularization strength
<code>sklearn.linear_model.SGDClassifier</code>	Mix and match loss and regularizer
<code>sklearn.linear_model.Perceptron</code>	SGD with perceptron loss and no regularization
<code>sklearn.svm.LinearSVC</code>	} Support Vector Machines - uses hinge loss SVC supports kernels
<code>sklearn.svm.SVC</code>	

Training sklearn models

```
from sklearn.linear_model import LogisticRegression
```

```
clf = LogisticRegression()  
clf.fit(X_train, y_train)
```

```
y_pred = clf.predict(X_test)
```

```
np.sum(y_test == y_pred) / N_test
```

Calculate accuracy

$$\vec{\theta}^*, b^* = \arg \min_{\vec{\theta}, b} R_N(\vec{\theta}) + \lambda \Omega(\vec{\theta})$$

$$h(\vec{x}; \vec{\theta}, b) = \sigma(\vec{\theta} \cdot \vec{x} + b)$$

Practical Notes

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

```
class sklearn.linear_model.LogisticRegression(penalty='l2', *, dual=False,
tol=0.0001, C=1.0, fit_intercept=True, intercept_scaling=1, class_weight=None,
random_state=None, solver='lbfgs', max_iter=100, multi_class='deprecated',
verbose=0, warm_start=False, n_jobs=None, l1_ratio=None) \[source\]
```

What we
saw in class

$$\frac{1}{N} \sum_{i=1}^N \text{loss}_{\log}(y^{(i)}, \hat{y}^{(i)}) + \lambda \Omega(\vec{\theta})$$

What sklearn uses

$$C \sum_{i=1}^N \text{loss}_{\log}(y^{(i)}, \hat{y}^{(i)}) + \Omega(\vec{\theta})$$

Not every solver supports
every regularization penalty

solver	penalty
'lbfgs'	'l2', None
'liblinear'	'l1', 'l2'
'newton-cg'	'l2', None
'newton-cholesky'	'l2', None
'sag'	'l2', None
'saga'	'elasticnet', 'l1', 'l2', None

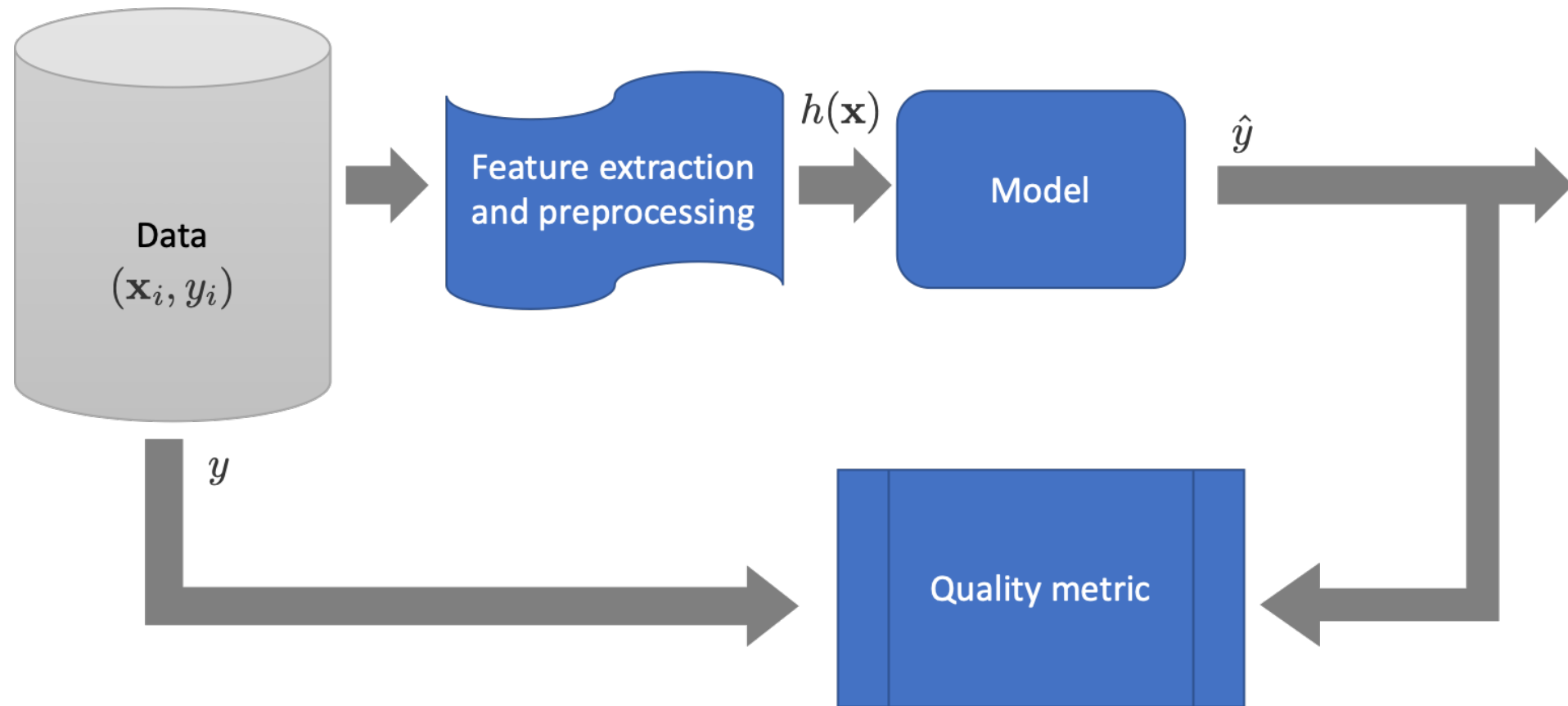
Model Assessment and Model Selection

- **Model assessment:** evaluating a model's performance
- **Model selection:** selecting the proper level of flexibility for a model (e.g. regularization strength in logistic regression)

“Remember that all models are wrong; the practical question is how wrong do they have to be to not be useful.”

— George Box, 1987

Evaluating Model Performance



Model Assessment

- Performance Metrics:
 - Classification: accuracy, precision, recall, sensitivity, specificity, true positive rate, false positive rate, F1-score, AUROC, AUPRC, calibration*
 - Regression: MSE, RMSE, MAE*, MBE*
- Process:
 - training/test split (holdout)
 - K-fold cross-validation, Monte Carlo cross validation
 - bootstrap confidence intervals (bootstrap CIs)*

*: you're not responsible to know for the exam

Classification Performance

Given $\hat{y} = h(\vec{x}; \vec{\theta})$ and $D = \{\vec{x}^{(i)}, y^{(i)}\}_{i=1}^N$ $\vec{x}^{(i)} \in \mathbb{R}^d$ $y^{(i)} \in \{-1, +1\}$

Misclassification error: $\frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}^{(i)} \neq y^{(i)}]$ Accuracy: $\frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}^{(i)} = y^{(i)}]$

Predicted

Actual

Correct?

1.

Not fraud

Not fraud



2.

Not fraud

Not fraud



3.

Not fraud

Fraud

X

4.

Fraud

Fraud



...

n.

Fraud

Not fraud

X



Classification Performance

Given $\hat{y} = h(\vec{x}; \vec{\theta})$ and $D = \{\vec{x}^{(i)}, y^{(i)}\}_{i=1}^N$ $\vec{x}^{(i)} \in \mathbb{R}^d$ $y^{(i)} \in \{-1, +1\}$

Misclassification error: $\frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}^{(i)} \neq y^{(i)}]$ Accuracy: $\frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}^{(i)} = y^{(i)}]$

What's wrong with accuracy?

- Example: predicting rare disease
- Can be inflated when data is imbalanced
 - Accuracy of the base case (predicting dominant class) can be very high
- Assumes equal cost for both types of “error”

Predicted

Actual

1.

Not fraud

Not fraud

2.

Not fraud

Not fraud

3.

Not fraud

Fraud

4.

Fraud

Fraud

n.

Fraud

Not fraud

False
Negative

False
Positive



Predicted

Actual

1.

Not fraud

Not fraud

2.

Not fraud

Not fraud

3.

Not fraud

Fraud

4.

Fraud

Fraud

n.

Fraud

Not fraud

True
Negative

True
Positive



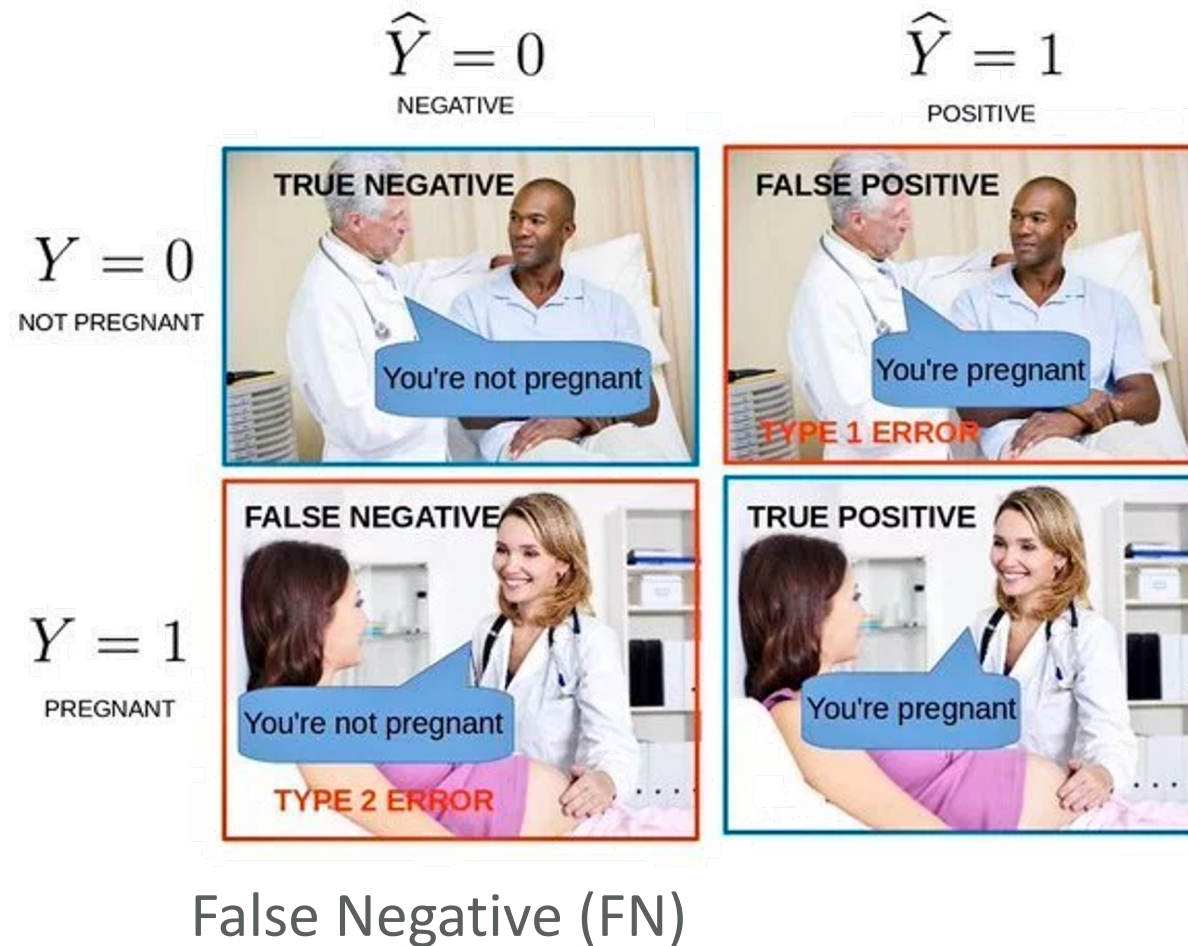
Confusion Matrix (for binary classification)

	Predicted: (−)	Predicted: (+)
Actual: (−)	True Negative (TN)	False Positive (FP)
Actual: (+)	False Negative (FN)	True Positive (TP)

“Table of confusion”

https://en.wikipedia.org/wiki/Confusion_matrix

Confusion Matrix



Classification Metrics

	Pred: (-)	Pred: (+)
Actual: (-)	TN	FP
Actual: (+)	FN	TP

$(TP+TN)/N$ accuracy

$TP/(TP+FN)$ true positive rate (TPR), sensitivity, recall

$TN/(TN+FP)$ true negative rate (TNR), specificity

$FP/(TN+FP)$ false positive rate (FPR), $1 - \text{specificity}$

$TP/(TP+FP)$ precision, positive predictive value (PPV)

What's the range of each metric (min and max possible value)?

What if I predict everything positive? Or everything negative?

Can we optimize all metrics at the same time?

Composite Metrics: Trade-offs

	Pred: (-)	Pred: (+)
Actual: (-)	TN	FP
Actual: (+)	FN	TP

Balanced Accuracy: mean of **sensitivity** and **specificity**

$$\frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right)$$

F1-score: harmonic mean of **precision** and **recall**

$$\begin{aligned} F_1 &= \frac{2}{\frac{1}{TPR} + \frac{1}{PPV}} \\ &= 2 \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

$$F_\beta = (1 + \beta^2) \frac{\text{Precision} * \text{Recall}}{(\beta^2 \text{Precision}) + \text{Recall}}, \beta > 0$$

Discrimination Thresholds

Predicted Class $\{-1, +1\}$

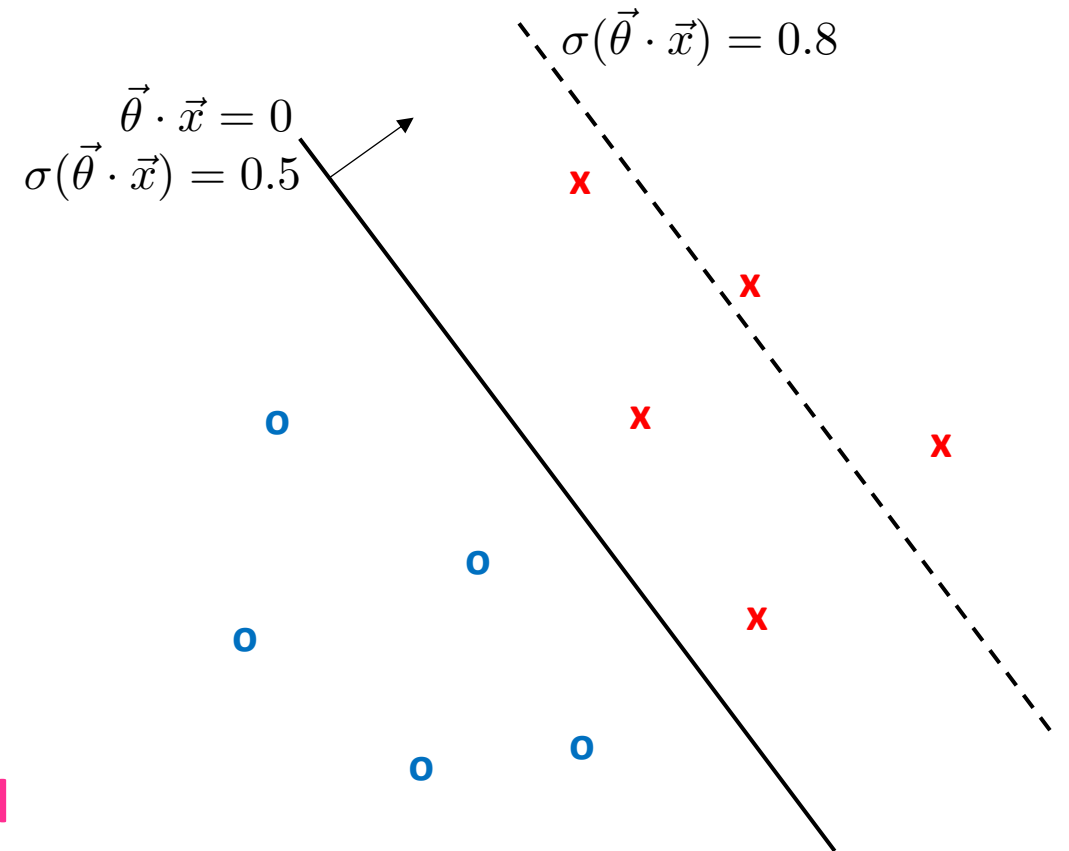
$$h(\vec{x}; \vec{\theta}) = \text{sign}(\vec{\theta} \cdot \vec{x})$$

Predicted Class Probability $[0,1]$

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

Normally, we output positive label
if probability > 0.5 or $\vec{\theta} \cdot \vec{x} > 0$

**What if we use a different threshold based
on the probability to make class prediction?**

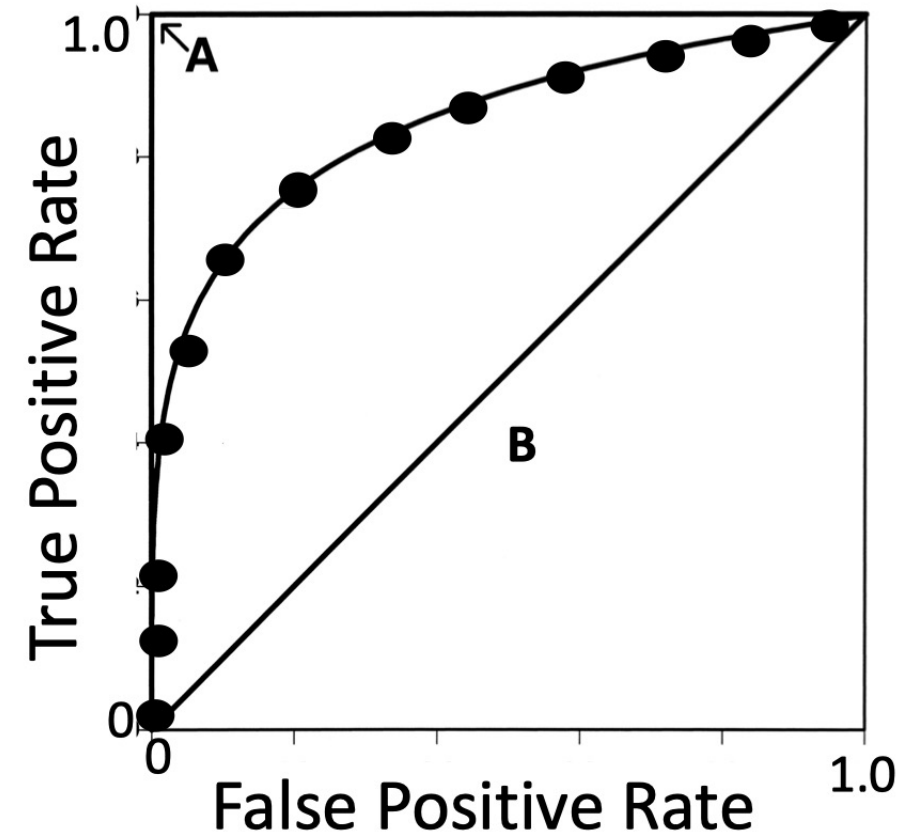


Receiver Operating Characteristic Curve

- Y-axis: true positive rate (sensitivity)
- X-axis: false positive rate (1-specificity)

How to get multiple points?

- Each point on the ROC curve corresponds to a threshold / a decision boundary
- Each point represents a different tradeoff between FPR and TPR



ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

$$\hat{y}^{(1)} = 90\% \quad y^{(1)} = +1$$

$$\hat{y}^{(2)} = 80\% \quad y^{(2)} = +1$$

$$\hat{y}^{(3)} = 70\% \quad y^{(3)} = -1$$

$$\hat{y}^{(4)} = 60\% \quad y^{(4)} = +1$$

$$\hat{y}^{(5)} = 40\% \quad y^{(5)} = -1$$

$$\hat{y}^{(6)} = 30\% \quad y^{(6)} = -1$$

$$\hat{y}^{(7)} = 20\% \quad y^{(7)} = +1$$

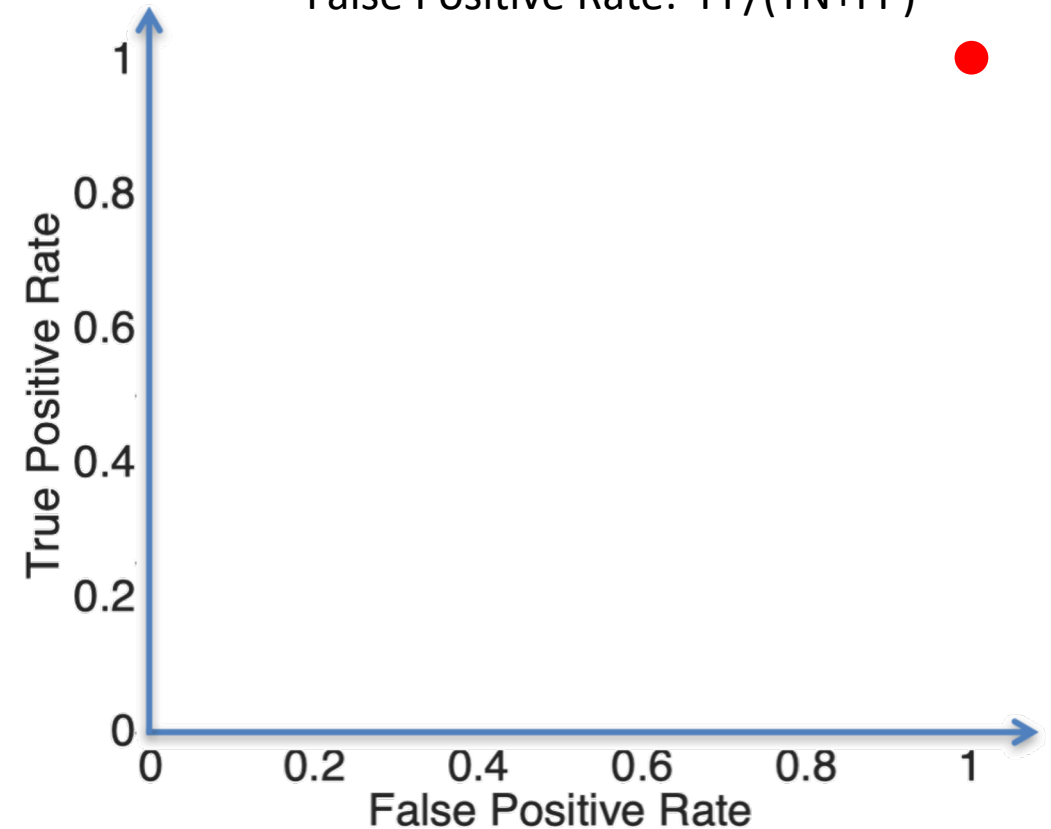
$$\hat{y}^{(8)} = 10\% \quad y^{(8)} = -1$$

Pos.

Neg.

True Positive Rate: $TP/(TP+FN)$

False Positive Rate: $FP/(TN+FP)$



ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

$$\hat{y}^{(1)} = 90\%$$

$$y^{(1)} = +1$$

$$\hat{y}^{(2)} = 80\%$$

$$y^{(2)} = +1$$

$$\hat{y}^{(3)} = 70\%$$

$$y^{(3)} = -1$$

$$\hat{y}^{(4)} = 60\%$$

$$y^{(4)} = +1$$

$$\hat{y}^{(5)} = 40\%$$

$$y^{(5)} = -1$$

$$\hat{y}^{(6)} = 30\%$$

$$y^{(6)} = -1$$

$$\hat{y}^{(7)} = 20\%$$

$$y^{(7)} = +1$$

$$\hat{y}^{(8)} = 10\%$$

$$y^{(8)} = -1$$

Pos.

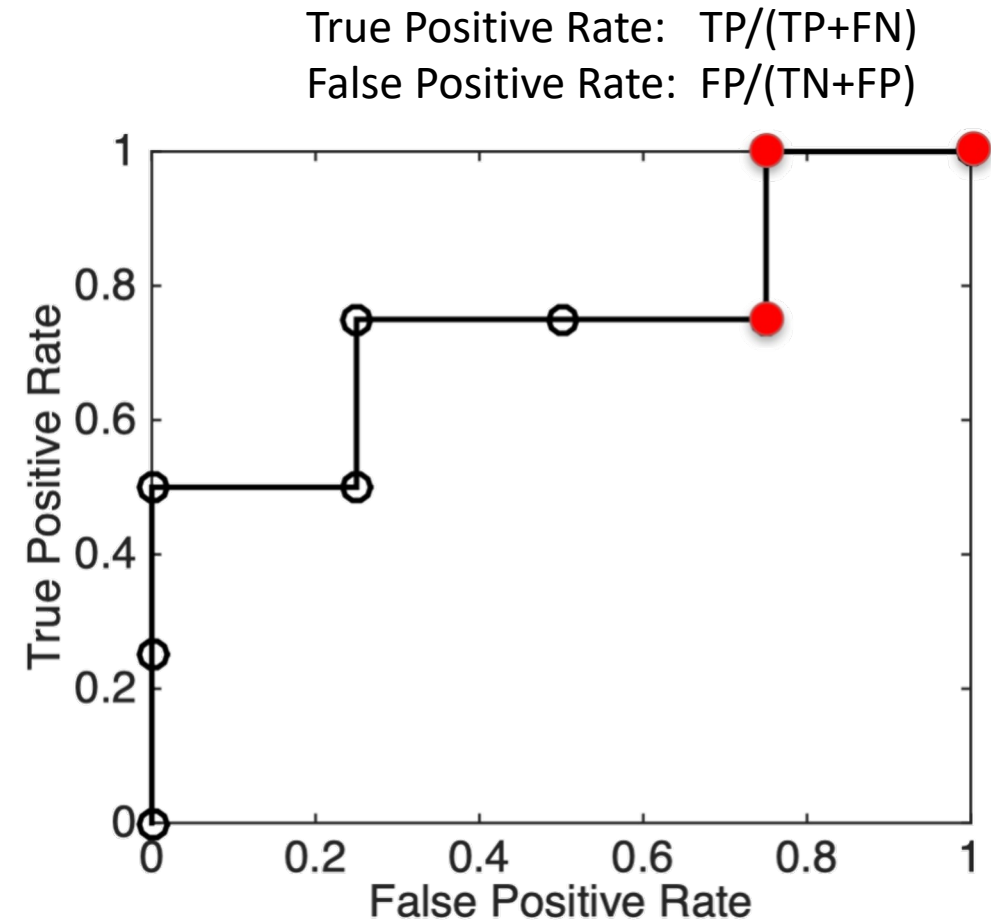
Neg.



ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

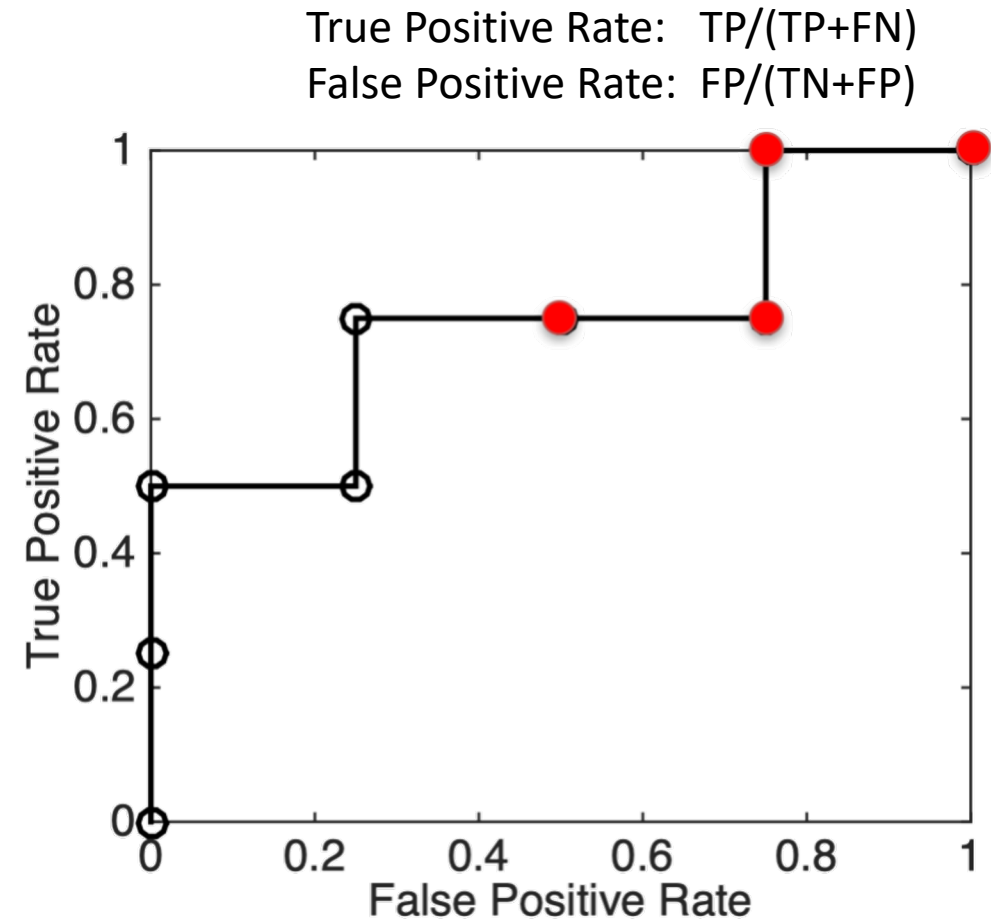
$\hat{y}^{(1)} = 90\%$	$y^{(1)} = +1$	Pos.
$\hat{y}^{(2)} = 80\%$	$y^{(2)} = +1$	
$\hat{y}^{(3)} = 70\%$	$y^{(3)} = -1$	
$\hat{y}^{(4)} = 60\%$	$y^{(4)} = +1$	
$\hat{y}^{(5)} = 40\%$	$y^{(5)} = -1$	
$\hat{y}^{(6)} = 30\%$	$y^{(6)} = -1$	
$\hat{y}^{(7)} = 20\%$	$y^{(7)} = +1$	Neg.
$\hat{y}^{(8)} = 10\%$	$y^{(8)} = -1$	



ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

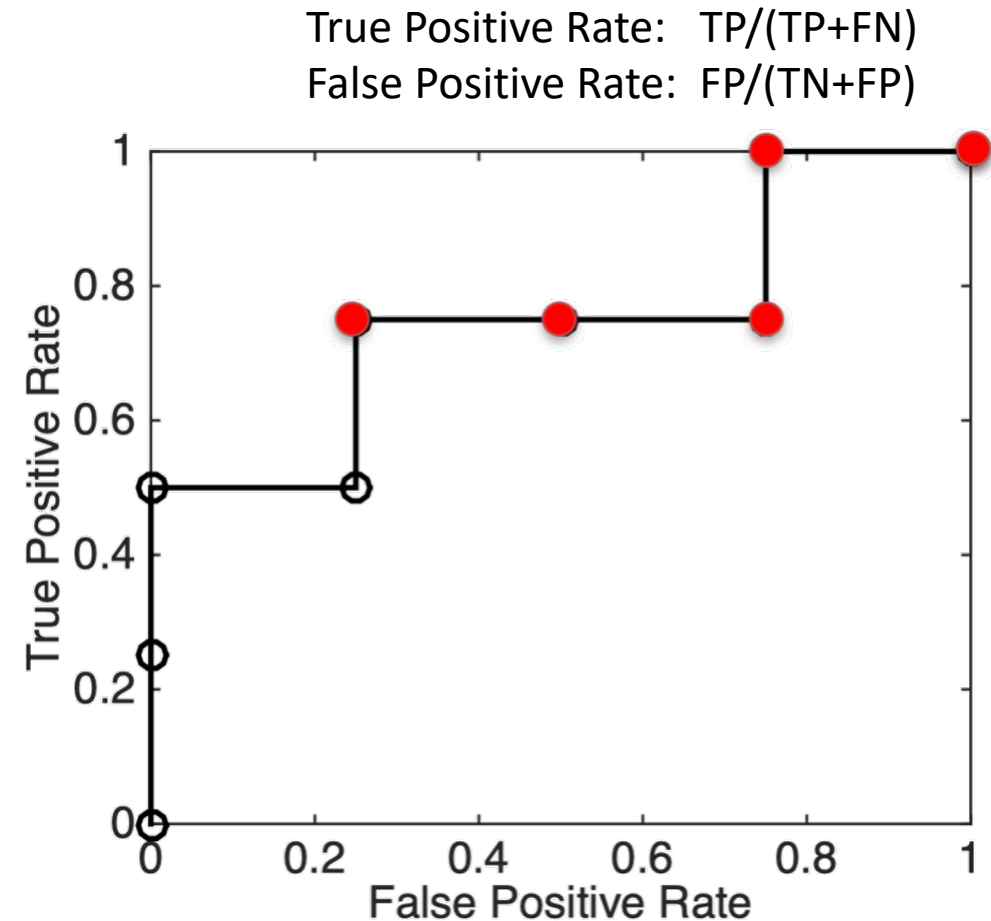
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$\hat{y}^{(3)} = 70\%$	$y^{(3)} = -1$	
$\hat{y}^{(4)} = 60\%$	$y^{(4)} = +1$	
$\hat{y}^{(5)} = 40\%$	$y^{(5)} = -1$	
		Neg.
$\hat{y}^{(6)} = 30\%$	$y^{(6)} = -1$	
$\hat{y}^{(7)} = 20\%$	$y^{(7)} = +1$	
$\hat{y}^{(8)} = 10\%$	$y^{(8)} = -1$	



ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

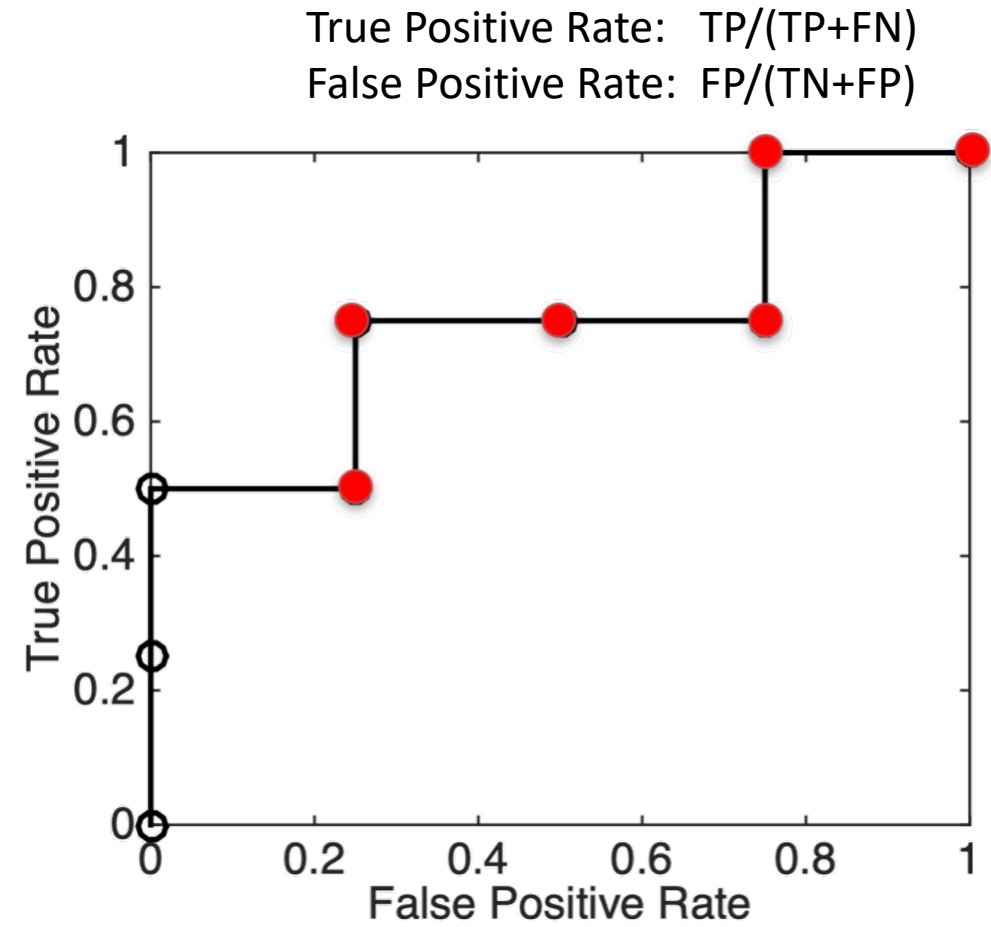
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$\hat{y}^{(2)} = 80\%$	$y^{(2)} = +1$	
$\hat{y}^{(3)} = 70\%$	$y^{(3)} = -1$	
$\hat{y}^{(4)} = 60\%$	$y^{(4)} = +1$	Pos.
$\hat{y}^{(5)} = 40\%$	$y^{(5)} = -1$	Neg.
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ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

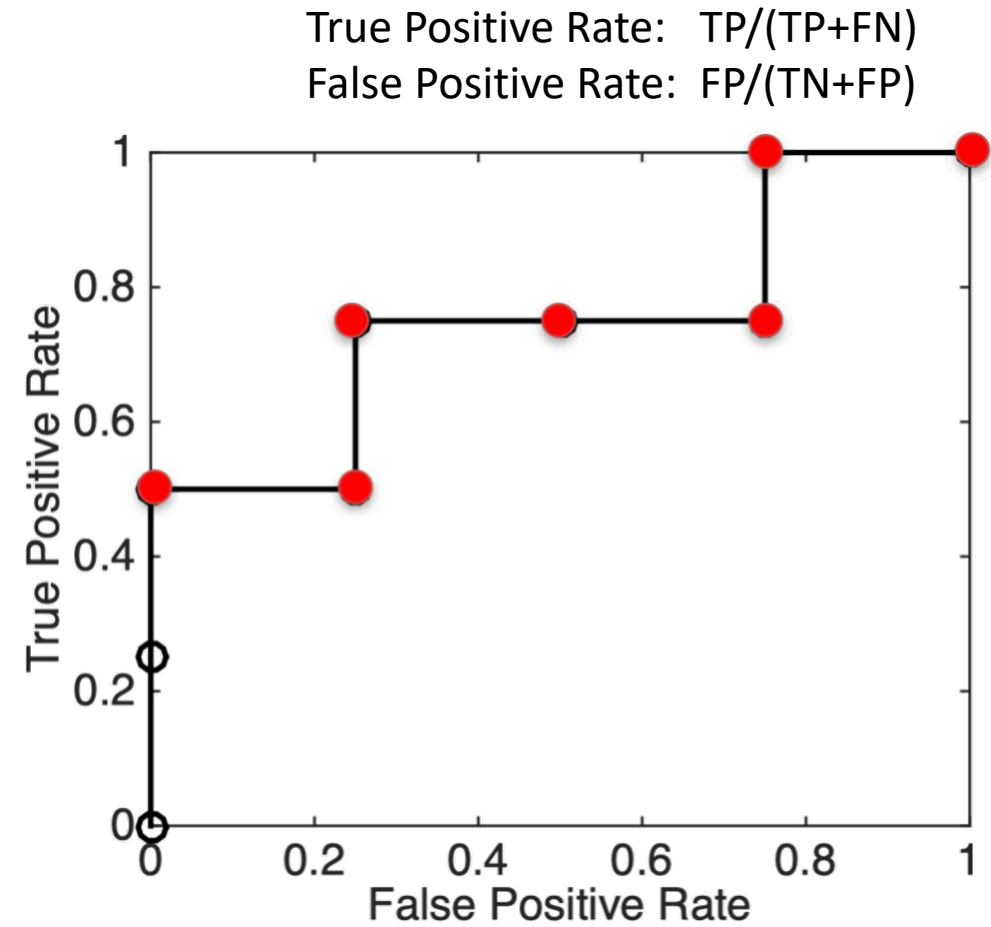
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$\hat{y}^{(7)} = 20\%$	$y^{(7)} = +1$	
$\hat{y}^{(8)} = 10\%$	$y^{(8)} = -1$	



ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

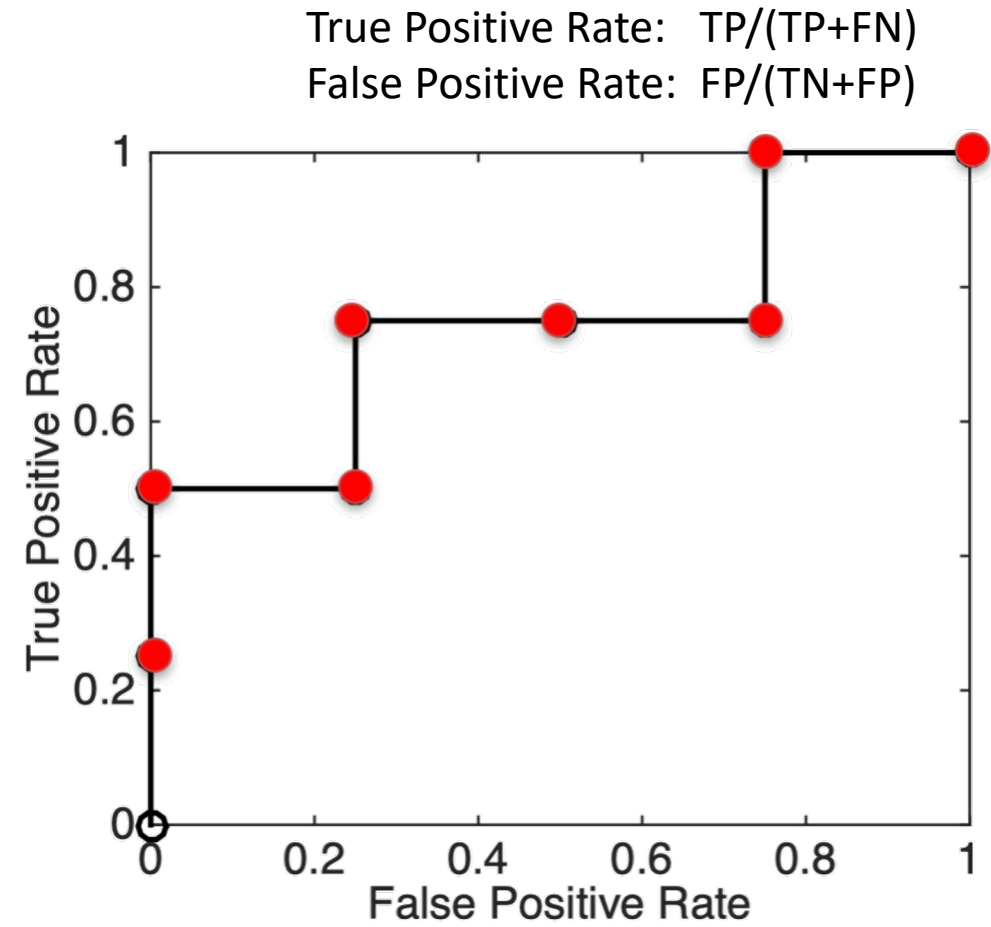
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$\hat{y}^{(5)} = 40\%$	$y^{(5)} = -1$	
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ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

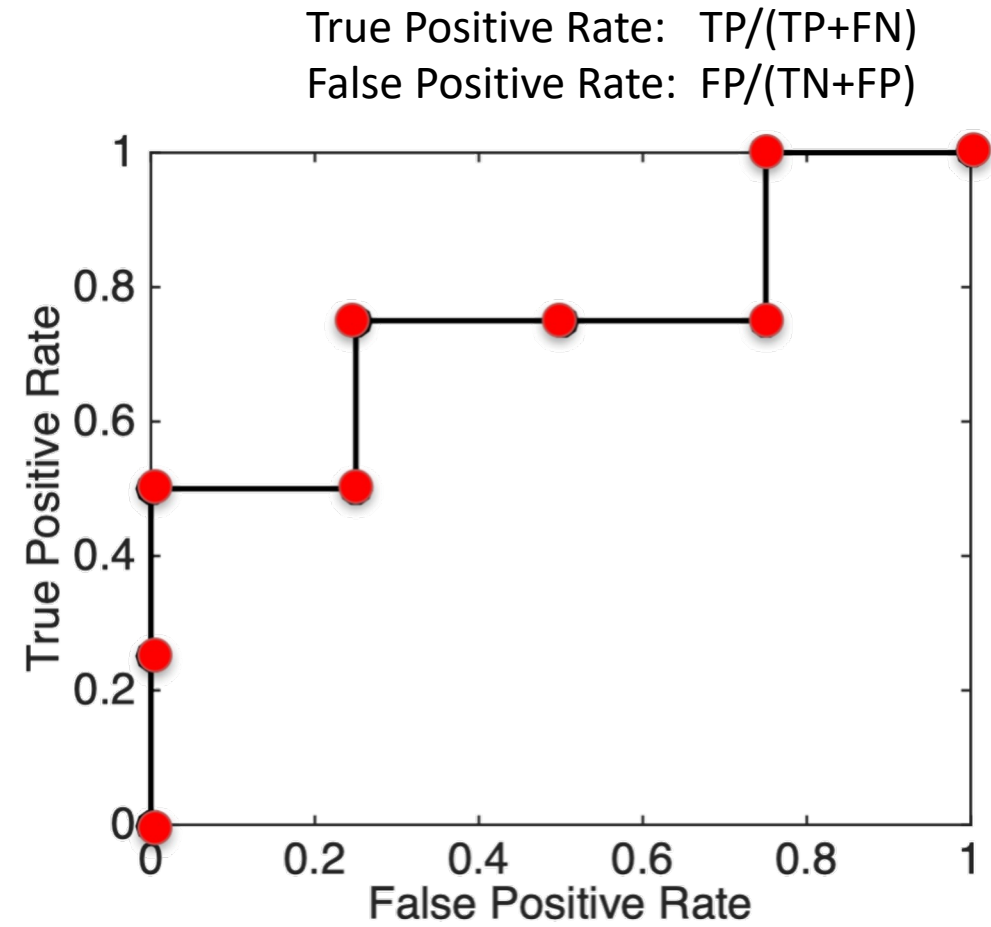
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$\hat{y}^{(5)} = 40\%$	$y^{(5)} = -1$	
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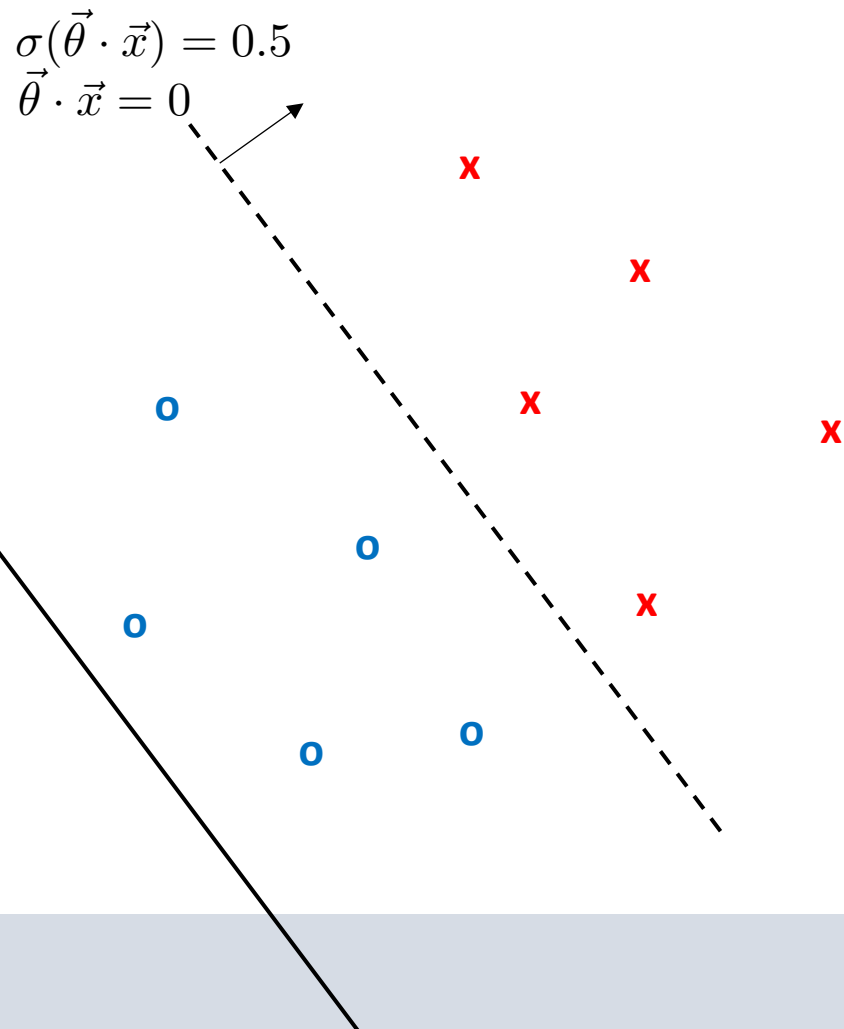
ROC curve: How to get multiple points?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

		Pos. Neg.
$\hat{y}^{(1)} = 90\%$	$y^{(1)} = +1$	
$\hat{y}^{(2)} = 80\%$	$y^{(2)} = +1$	
$\hat{y}^{(3)} = 70\%$	$y^{(3)} = -1$	
$\hat{y}^{(4)} = 60\%$	$y^{(4)} = +1$	
$\hat{y}^{(5)} = 40\%$	$y^{(5)} = -1$	
$\hat{y}^{(6)} = 30\%$	$y^{(6)} = -1$	
$\hat{y}^{(7)} = 20\%$	$y^{(7)} = +1$	
$\hat{y}^{(8)} = 10\%$	$y^{(8)} = -1$	



ROC curve: How to get multiple points?



$$\hat{y}^{(1)} = 90\% \quad y^{(1)} = +1$$

$$\hat{y}^{(2)} = 80\% \quad y^{(2)} = +1$$

$$\hat{y}^{(3)} = 70\% \quad y^{(3)} = -1$$

$$\hat{y}^{(4)} = 60\% \quad y^{(4)} = +1$$

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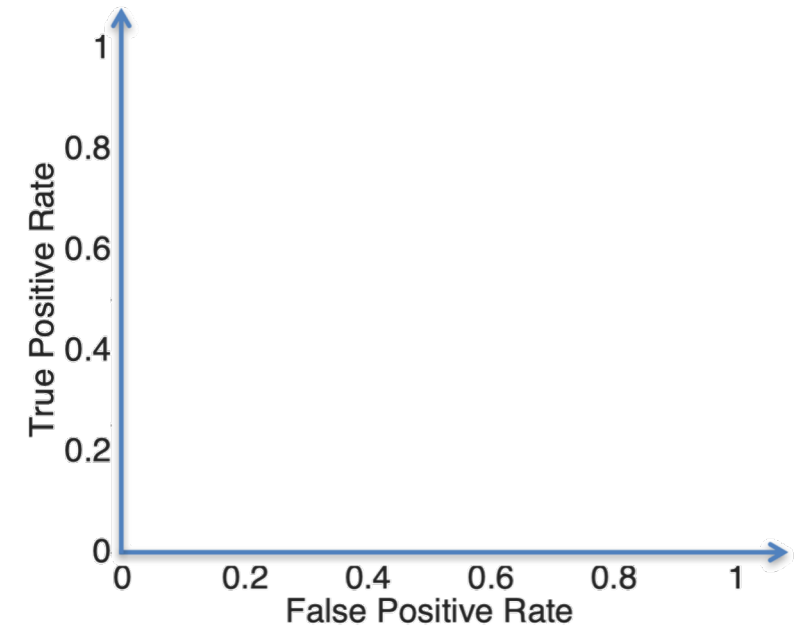
$$\hat{y}^{(6)} = 30\% \quad y^{(6)} = -1$$

$$\hat{y}^{(7)} = 20\% \quad y^{(7)} = +1$$

$$\hat{y}^{(8)} = 10\% \quad y^{(8)} = -1$$

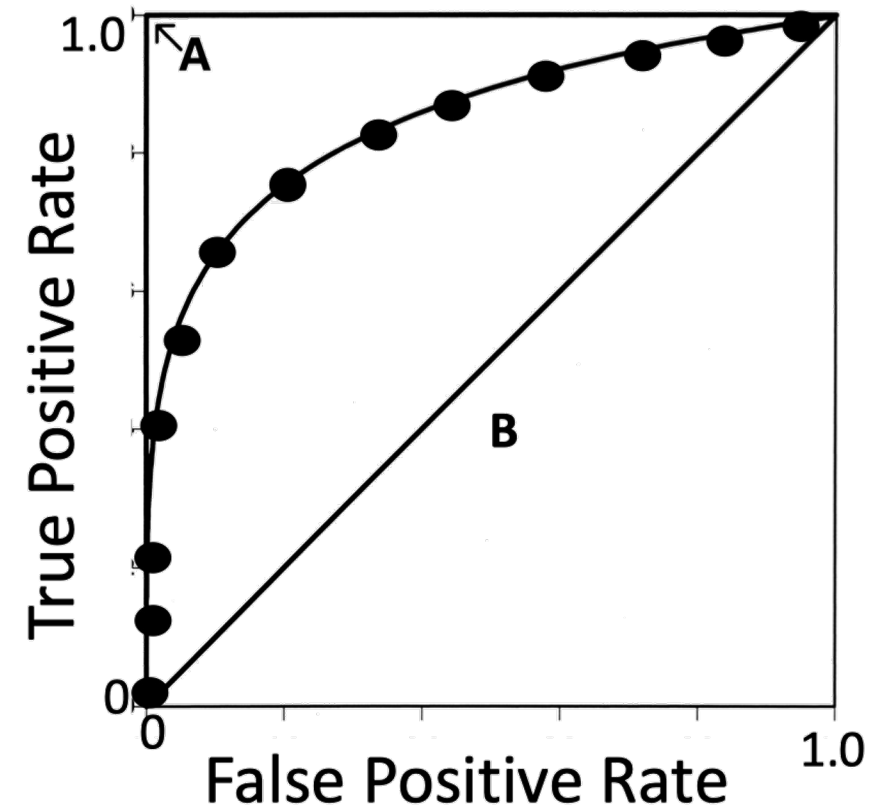
Pos.
Neg.

True Positive Rate: $TP/(TP+FN)$
False Positive Rate: $FP/(TN+FP)$



ROC Curve

- Curve is always increasing (slope ≥ 0)
- Two non-intersecting curves means one model dominates the other
- A: perfect predictions
- B: random predictions
- Shows the trade-off between sensitivity and specificity
- But still not a very good summary metric, it's not a single number



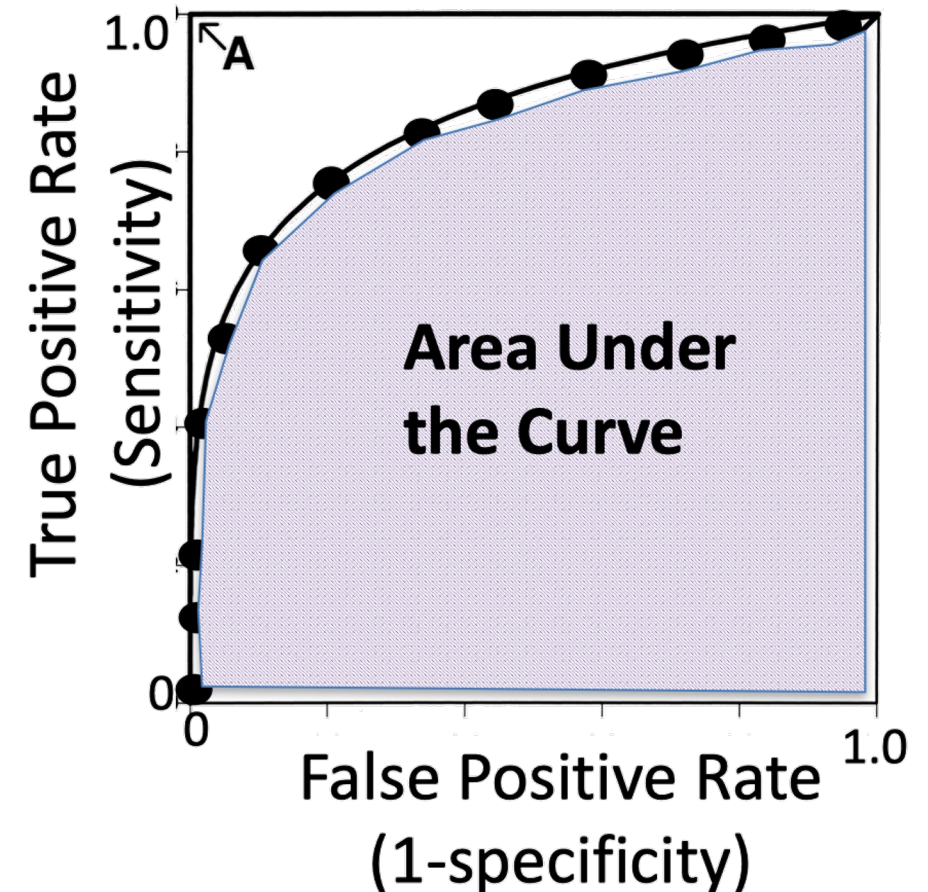
Area Under ROC Curve (AUROC, ROC-AUC)

Calculated using the trapezoid rule
`sklearn.metrics.auc(x, y)`

Intuitive meaning:

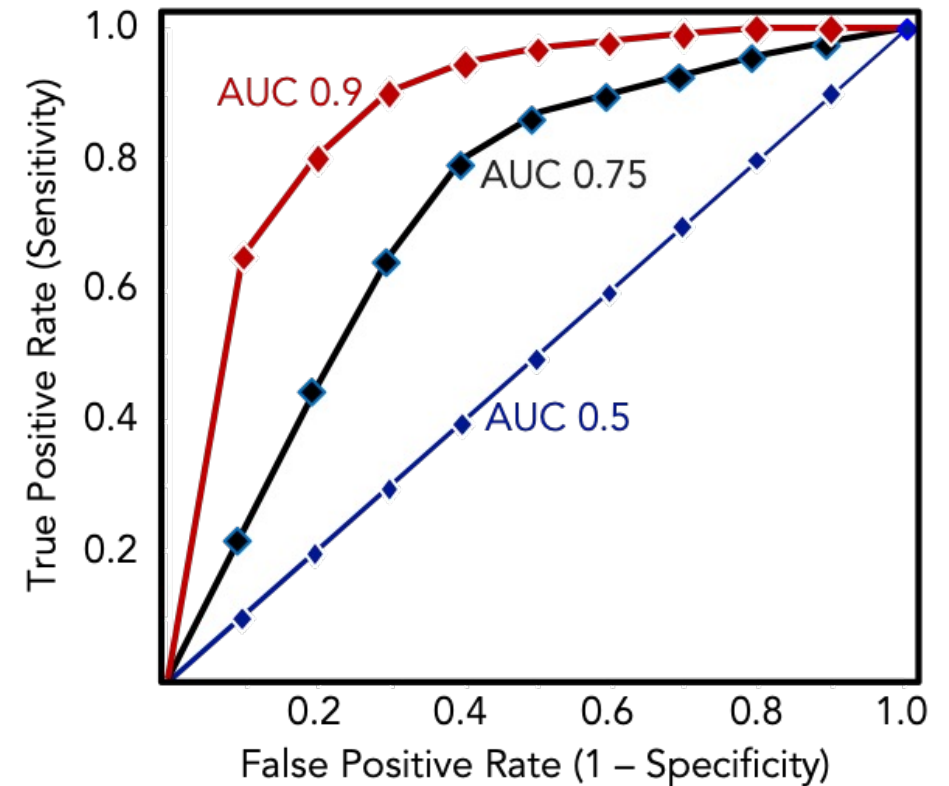
- Given two randomly chosen examples, one pos. and one neg., prob. of ranking pos. example higher

$$\text{AUROC} = P[\text{score}(x^+) > \text{score}(x^-)]$$



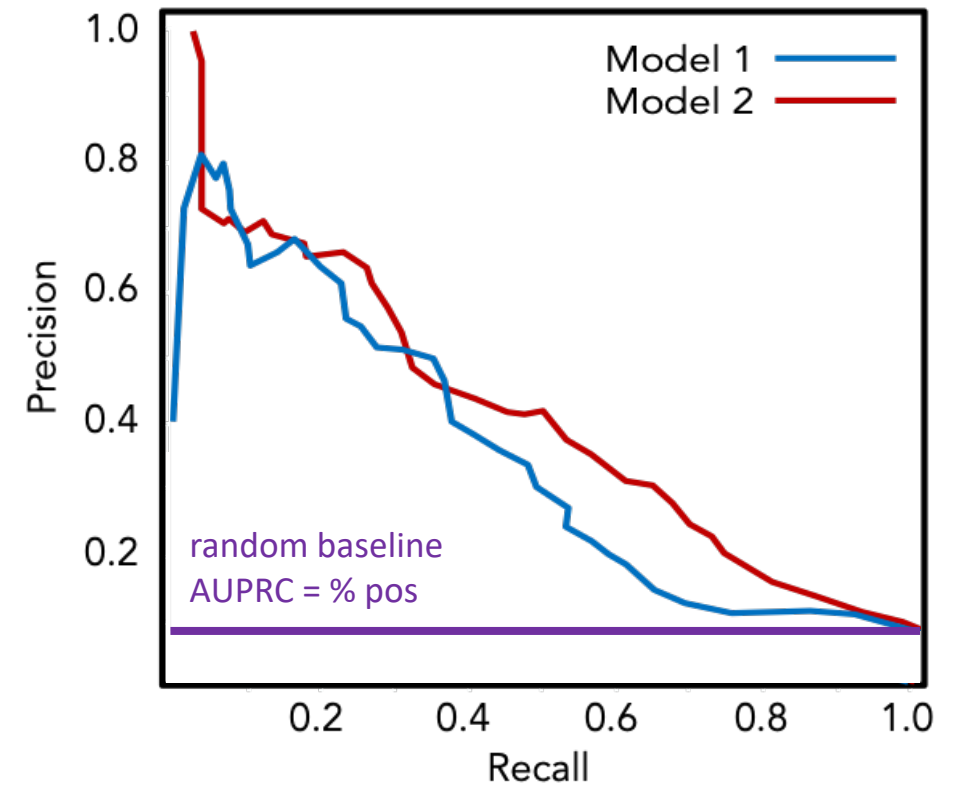
AUROC

- 1: perfect prediction
- > 0.9 : excellent prediction — something potentially fishy, should check for information leakage
- 0.8: good prediction
- 0.5: random prediction
- < 0.5 : something wrong!



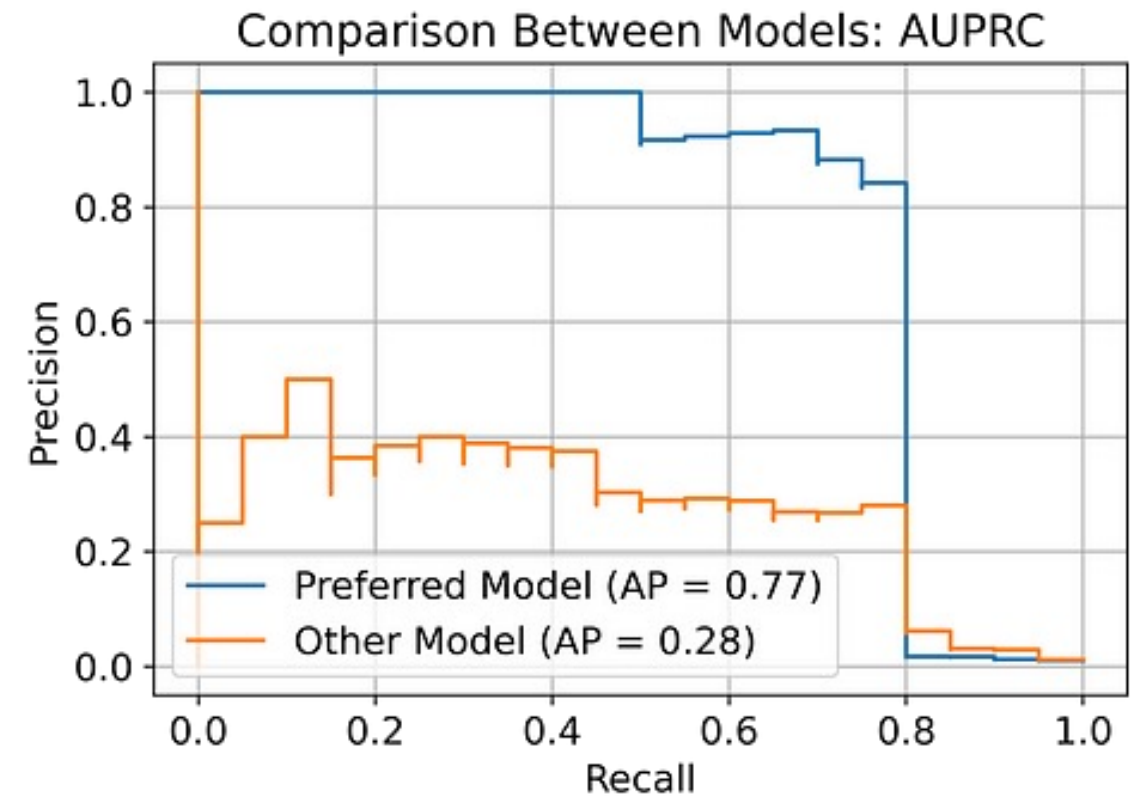
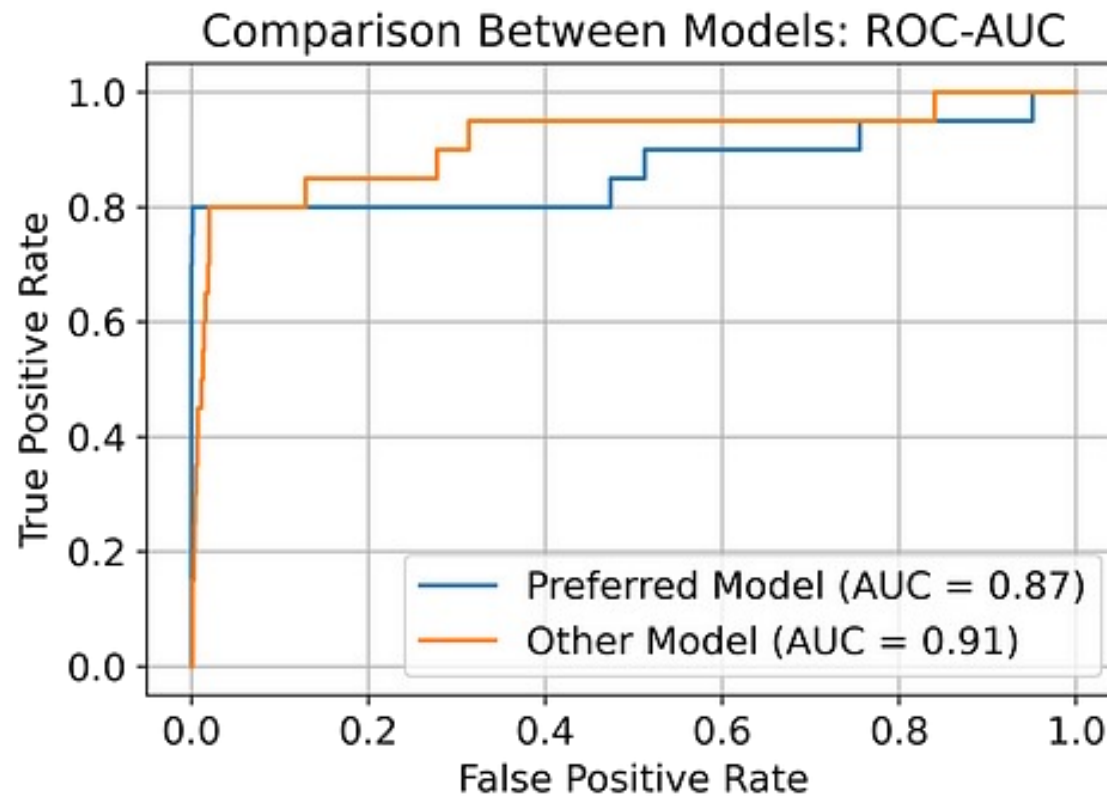
Precision-Recall Curve & AUPRC

- PR curve can be constructed in a similar manner by sweeping classification threshold
- A high AUPRC represents both high recall and precision
- In practice, recommend to use both



AUROC vs. AUPRC (imbalanced data)

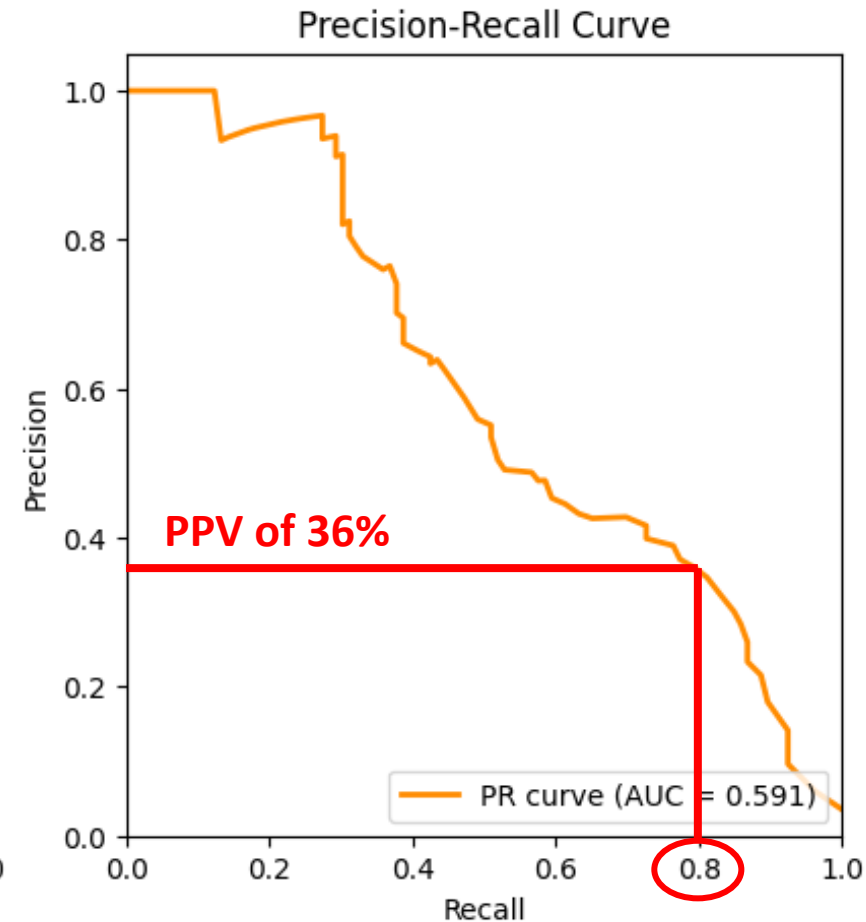
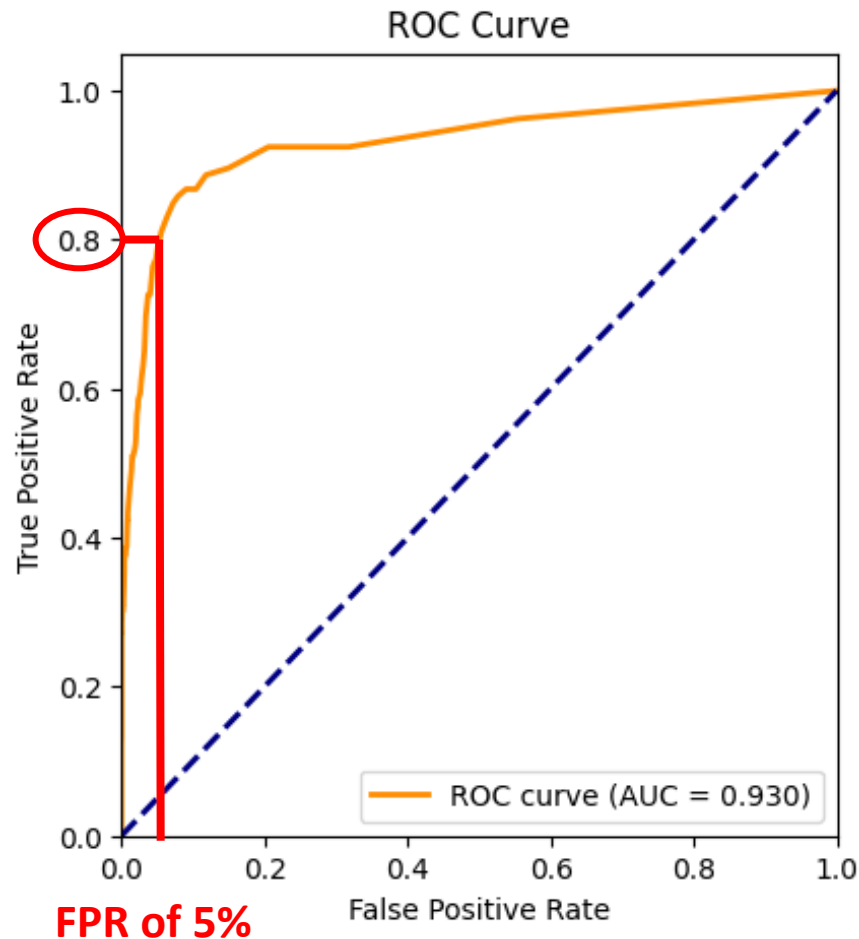
20 positives; 2000 negatives



<https://towardsdatascience.com/imbalanced-data-stop-using-roc-auc-and-use-auprc-instead-46af4910a494>

Selecting a Threshold

e.g. “I want a minimum sensitivity of 80%”
→ makes sure we catch 80% of the positive cases



*Classifier Probability Calibration

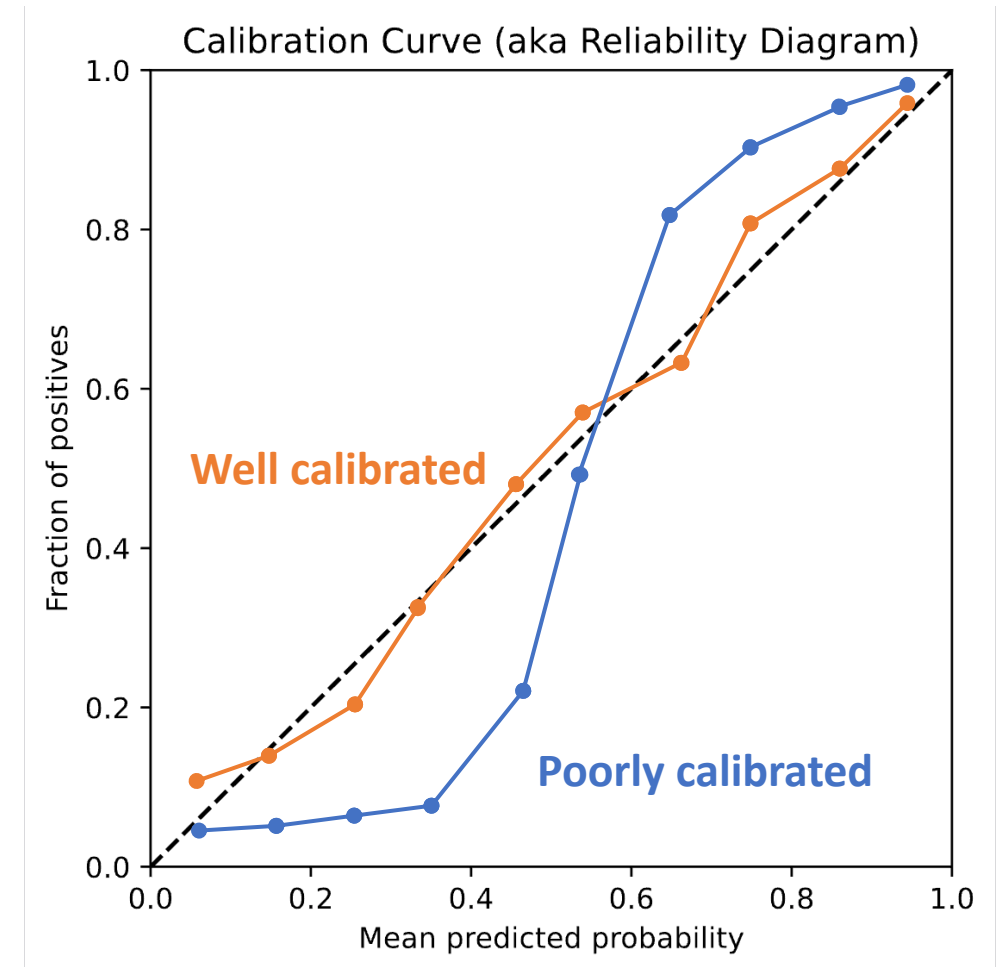
When a model produce a probability of positive class, is that number actually a meaningful probability?

$$h(\vec{x}; \vec{\theta}) = \sigma(\vec{\theta} \cdot \vec{x})$$

If my model predicts 90% for a set of examples, does that mean 90% of those examples have label +1?

A prediction of “90% chance of spam” should be spam 90% of the time.

<https://scikit-learn.org/stable/modules/calibration.html>



Metrics for Multiple Classes

- Accuracy:

$$ACC = \frac{TP1 + TP2 + TP3}{\text{Total}}$$

- Macro-average precision:

$$PRE_{\text{macro}} = \frac{PRE_1 + PRE_2 + PRE_3}{3}$$

- Micro-average precision:

$$PRE_{\text{micro}} = \frac{TP1 + TP2 + TP3}{TP1 + TP2 + TP3 + FP1 + FP2 + FP3}$$

(micro-average precision = micro-recall = accuracy)

	Predicted Class 1	Predicted Class 2	Predicted Class 3
Actual Class 1	TP1		
Actual Class 2		TP2	
Actual Class 3			TP3

Regression Metrics

Given $\hat{y} = f(\vec{x}; \vec{\theta})$ and $D = \{\vec{x}^{(i)}, y^{(i)}\}_{i=1}^N$ $\vec{x}^{(i)} \in \mathbb{R}^d$ $y^{(i)} \in \mathbb{R}$

Mean squared error

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \left(\hat{y}^{(i)} - y^{(i)} \right)^2$$

Mean absolute error

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N \left| \hat{y}^{(i)} - y^{(i)} \right|$$

Root mean squared error

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\hat{y}^{(i)} - y^{(i)} \right)^2}$$

Mean bias error

$$\text{MBE} = \frac{1}{N} \sum_{i=1}^N \left(\hat{y}^{(i)} - y^{(i)} \right)$$

For each metric, think about the following:

- What is its range (minimum and maximum possible score)?
- What's the “perfect score”?